Course 10

Software process quality

ISO – CMM Differences

ISO9001:2000	CMMI-DEV
International standard, applies to all types of organizations, supports both product and service oriented organizations	Written specifically for software development companies
A brief document – about 25 pages long, identifying the minimal requirements for a quality system	A detailed document – over 500 pages long
Emphasizes on a management of continuous improvement process, based on the PDCA (Plan-Do-Check-Act) model	Emphasizes on achieving "maturity" and improving its process continuously
One level of standard. The standard is based on recommendation	Defines 5 maturity levels of the organization, covering 25 process areas (PAs)

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ISO – CMM Differences

ISO 9000	SW-CMMI
Outwardly focused	Inwardly focused
Minimum requirements with implied continuous improvements	Explicit continuous quality improvement
Registration Document	No documentation

Certification audit for a 50 employee organization will be executed by 1 -12 auditors during one day	Certification audit for a 50 employee organization will be executed by 4 auditors during 4-5 days

Netta Dotan, Quality Assurance & project management, Ronkal Office Technologies

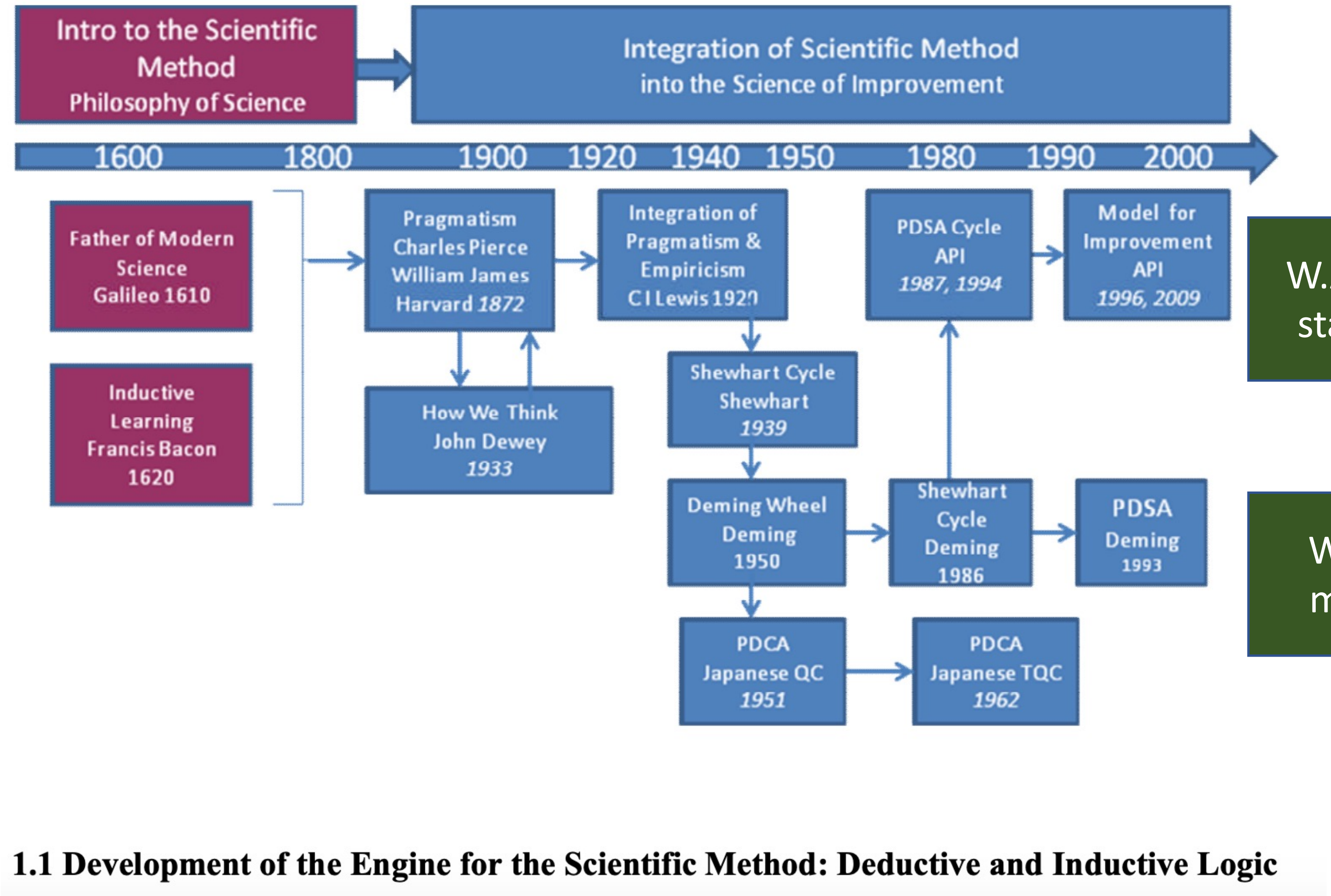
PDSA

- PDSA = Plan – Do – Study - Act

Former

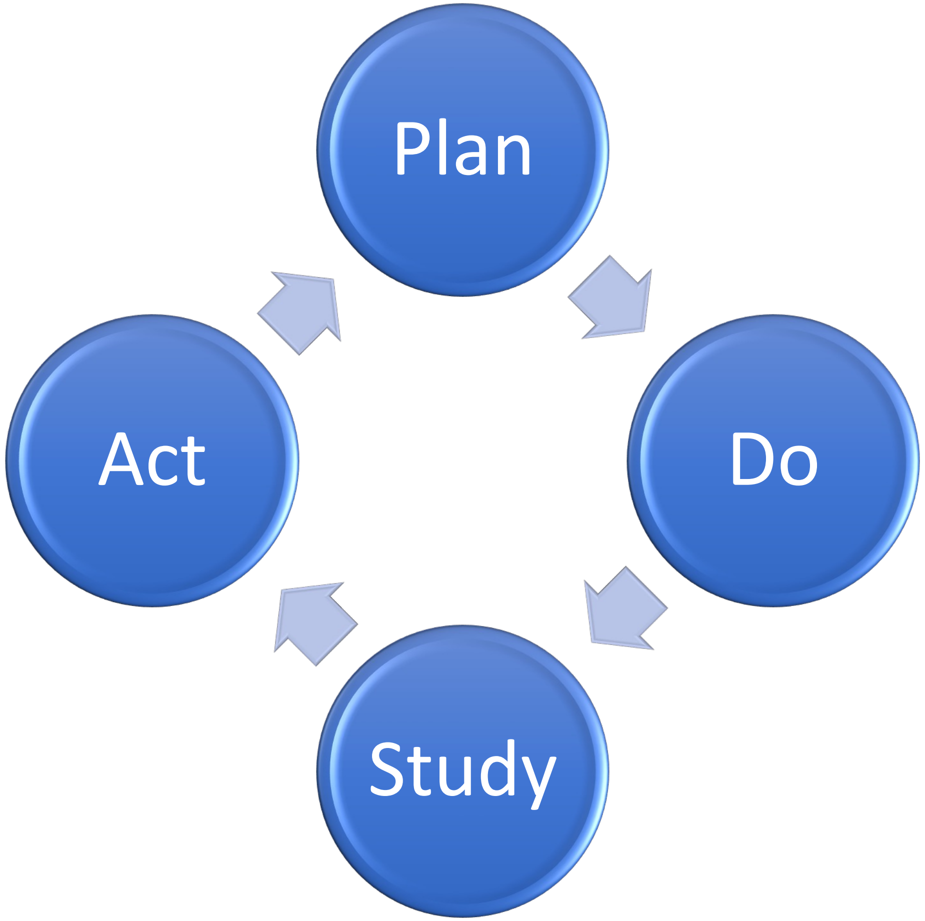
- PDCA = Plan – Do – Check – Act / Adjust

A little bit of history (R.Moen, C. Norman - Evolution of the PDCA Cycle)



W.A. Shewhart – father of statistical quality control

W.E. Deming– father of modern quality control



Iterative

Separation of phases

Plan

- Objectives
- Questions and predictions
- Plan to carry out

Do

- Carry out plan
- Document problems and unexpected observations
- Gather data
- Begin analysis of data

Study

- Complete analysis of data
- Compare data to the predictions
- Summarize conclusions
- How: gap analysis, appraisals

Act

- Find root of issues
- Re-evaluate risk
- Determine changes to de made
- Implement changes in order to prepare next iteration

Conclusions:

- PDCA + 7 basic tools (charts) = foundation for improvement (kaizen) in Japan (‘50)
- Toyota: „Building people before building cars”

Who need / use SQA?

SQA

Management

Provides visibility into the effectiveness and efficiency of the processes being used and the resulting product quality

Developers

Provides feedback to the individual projects on the efficiency and effectiveness of the processes that they are following so they can be improved at the project level

Software Engineers

Provides feedback to the SE group on the organizational processes they have facilitated in developing so they can be improved at the organizational level

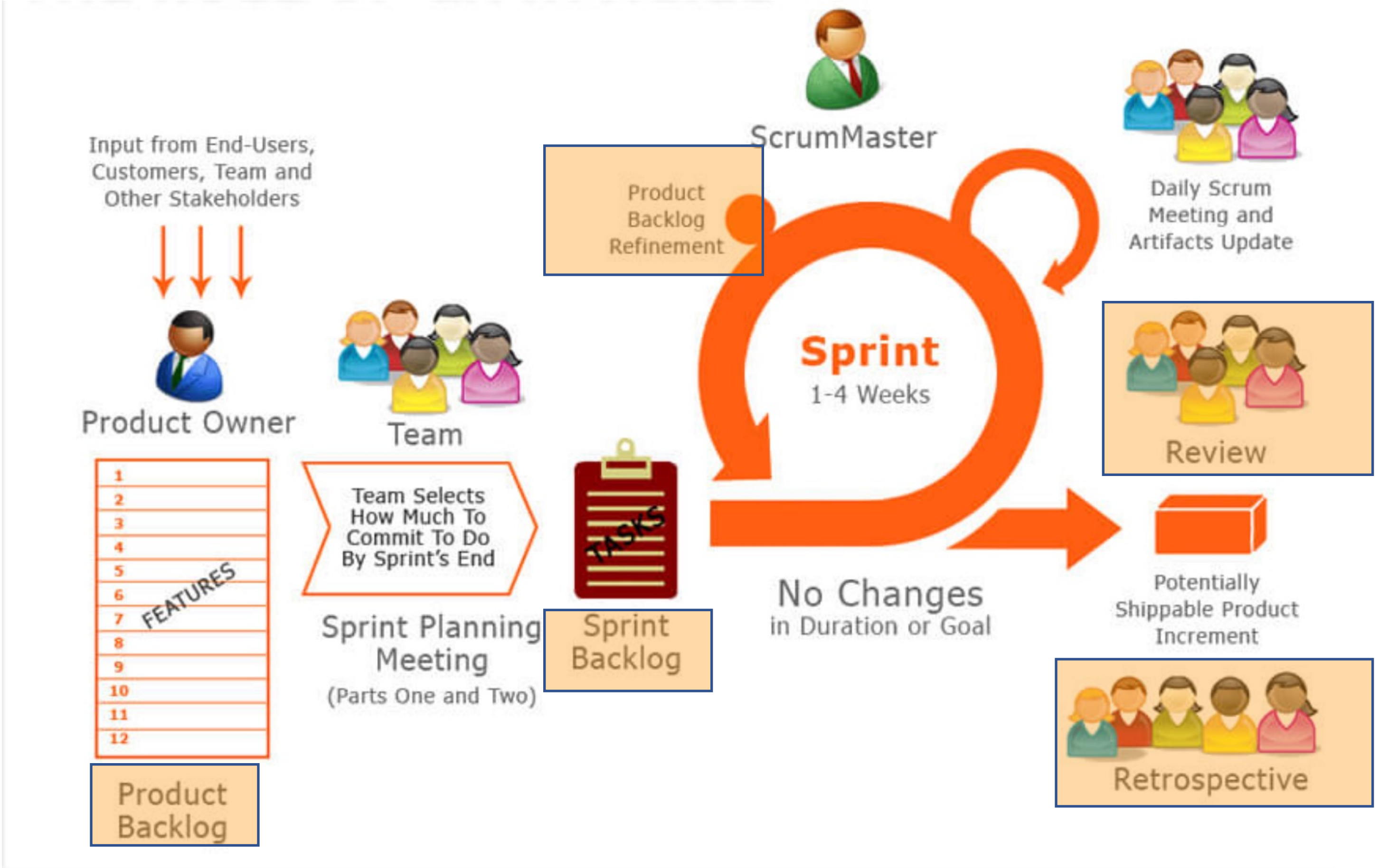
SQ and Agile

Agile methodologies

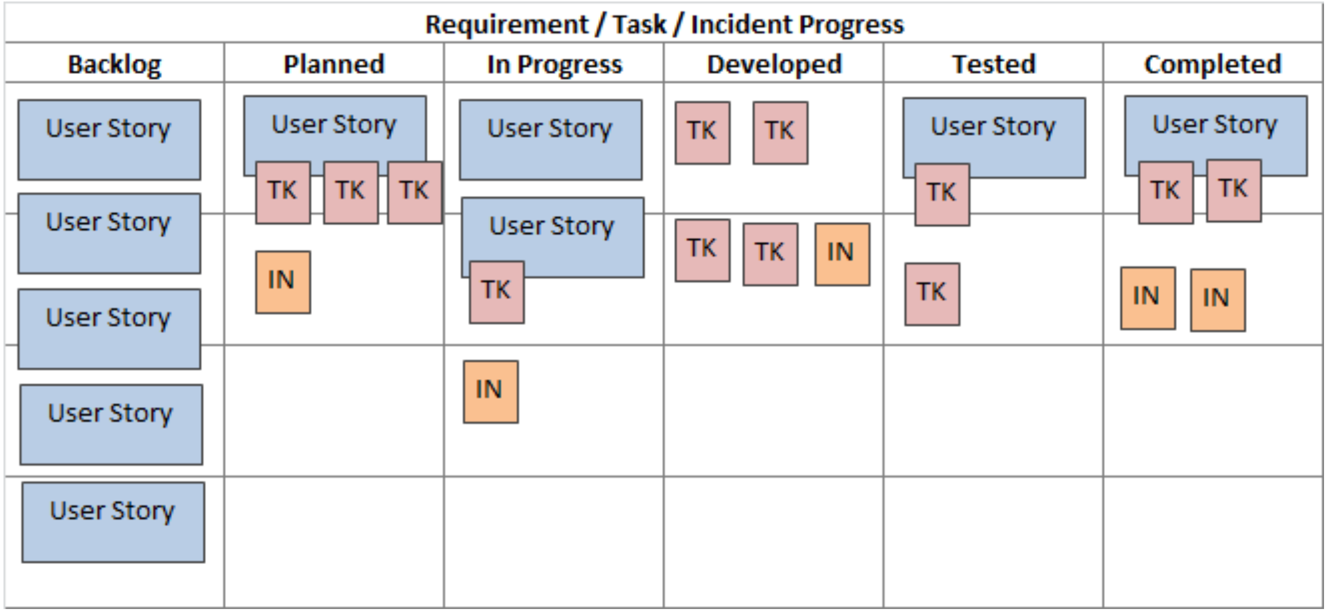
- **Scrum**
- Kanban
- Adaptive Software Development (ASD)
- Crystal Clear
- Dynamic Systems Development Method
- Extreme Programming
- Feature-Driven Development

Scrum – in short

[ref - figure]



Kanban – in short



- Visualize flows
- Limit work in progress
- aso

QA in SCRUM / Agile

- QA is not testing
- Internal auditing
- Responsible:
  - Detect waste
  - Find delays

=> Everyone responsible for QA

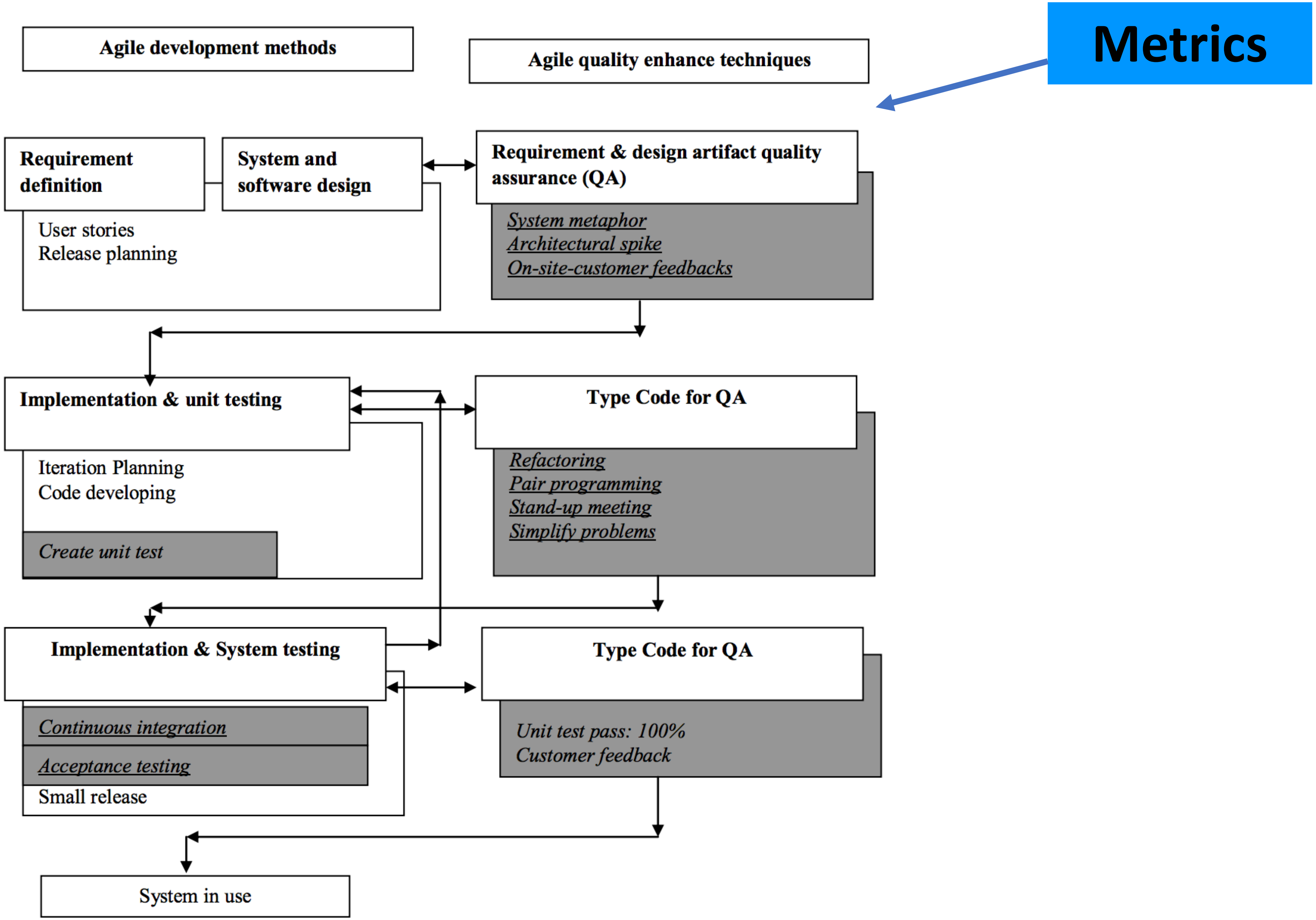
Olli P. Timperi - An Overview of Quality Assurance Practices in Agile Methodologies

TABLE IV
QUALITY ASSURANCE PRACTICES BY METHODOLOGY

Quality Assurance Practice	ASD	Crystal Clear	DSDM	FDD	Scrum	XP
Demonstration of software	X	X	X		X	
Joint application development	X		X	X		
Joint planning meeting		X			X	X
On-site customer		X	X		X	X
Prototyping			X			
Automated acceptance testing			X			X
Daily builds with testing		X		X	X	X
General testing	X	X	X	X	X	
Inspections	X			X	+	
Pair programming						X
Test-Driven development						X
Coding Standard		X	X	X		X
Collective code ownership						X
Personal code ownership				X	X	
Refactoring					+	X

X = the practice is used in the methodology

[A. Hossain , A. Kashem , S Sultana - Enhancing Software Quality Using Agile Techniques]



Key points

SQA in Agile:

- Continuous integration & testing ensure early defect detection
- Test-Driven Development (TDD) supports code quality from the start
- Retrospectives – should include quality concerns

Quality practices in Agile:

- Automation testing
- Definition of Done (DoD)
- Peer reviews & pair programming